Computer Networks Assignment

1. What are collision-free protocols? Why are they required in computer networks?

2. Explain the Basic Bit Map Protocol with a suitable diagram and example.

3. What is BRAP (Bit-Map Reservation Access Protocol)? Explain its basic working.

4. Explain the Binary Countdown Protocol with a suitable example.

5. What is MLMA (Multi-Level Multi-Access)? Explain its basic concept and working.

6. What are limited contention protocols? How do they differ from collision-free protocols?

7. Explain the Adaptive Tree Walk Protocol with a suitable diagram.

8. How does the Adaptive Tree Walk Protocol reduce collisions when multiple stations try to transmit data?

9. Compare collision-free protocols and limited contention protocols on the basis of working, collisions, efficiency, and applications.

10. What performance metrics are used to evaluate MAC protocols? Explain throughput, delay, channel utilization, and fairness.

11. What is the IEEE 802 family of standards? Explain its importance in computer networks.

12. Explain the IEEE 802.3 standard and its use in Ethernet networks.

13. Explain the IEEE 802.11 standard and its role in wireless LANs (Wi-Fi).

14. Write short notes on any four of the following IEEE standards: IEEE 802.1, 802.2, 802.3, 802.5, 802.11, and 802.15.

15. Differentiate between IEEE 802.3 (Ethernet) and IEEE 802.11 (Wi-Fi).

16. What is the Network Layer? Explain why it is required in computer networks.

17. Explain the major services provided by the Network Layer.

18. Describe the important design issues of the Network Layer.

19. Differentiate between connection-oriented and connectionless network services.

20. Explain the role of logical addressing and packet forwarding in the Network Layer.

21. What is routing? Explain the importance of routing algorithms in computer networks.

22. What is the Least Cost Routing Algorithm? Explain how the least-cost path between two nodes is selected.

23. Explain Dijkstra's Shortest Path Algorithm with its steps and a suitable example.

24. Consider a network with the following weighted links: A–B = 4, A–C = 2, B–C = 1, B–D = 5, C–D = 8, C–E = 10, and D–E = 2. Apply Dijkstra's algorithm to find the shortest paths from node A to all other nodes. Show the intermediate distance values.

25. Differentiate between the Least Cost Routing Algorithm and Dijkstra's Algorithm. Explain how Dijkstra's algorithm helps determine the least-cost paths in a network.